

Kade Bolen

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Education Overview

Boise State University

Boise, ID

Graduation: 12/2025

Boise State College of Engineering: Bachelor of Science for Computer Science

Technical Overview

Languages & Frameworks: Java, C, HTML, Javascript, Python, React, Flask, PostgreSQL

Tools: Git, GitHub, JSON, DataGrip, IntelliJ, Visual Studio Code, Draw.io, Jupyter Notebooks, CAD

Skills: Agile Development, Continuous Integration, Continuous Deployment, PostgreSQL Database Design, iOS Development, Windows Development, Web Application Development, Unit Testing, Object Oriented Programming, Containerization, 3D Object Design

Professional Overview

Albertsons Library Makerlab: Lead

Boise, ID

02/2024 – 12/2025

Oversee operations of the lab and ensure all colleagues feel secure and confident by providing assistance and expertise whenever needed.

- Designed standard operation procedures for 3D printers, laser cutters, CNC machines, and 3D scanners.
- Created proposal for a computer engineering capstone project on modifying existing machinery to more efficiently recycle plastic waste, a project which was recently completed.
- Spearheaded new operations within the lab such as plastic recycling, clay 3D printing, and custom equipment for a local physical rehabilitation hospital.
- Held colleagues accountable for their actions and made sure they remained on task while working as a team.
- Fixed hardware and software issues with equipment in the lab, e.g. mechanical failures with 3D printers, software connectivity issues with laser cutters, and firmware issues with custom 3D printers.

Albertsons Library Makerlab: Coach

Boise, ID

09/2023 – 02/2024

Teach patrons how to safely use equipment and support operations in the lab when needed.

- Helped lead tours and design workshops at the lab.
- Reported issues with equipment ASAP and see that they are fixed promptly.
- Assisted with community outreach.

Capstone Project

Genetic Rescue of Threatened Plant Species:

A full stack application complete with a **PostgreSQL database**, Rust back end with Diesel ORM, and a front facing website written in **HTML** and **Javascript**. The purpose of the application is to better understand patterns and find hidden correlations in years of handwritten data on threatened plant species, collected by scientists from Boise State University and Idaho National Laboratories. Our team adhered to the following practices;

- Consistent bi-weekly communication with our sponsors
- Continuous integration and deployment during development
- Thorough weekly sprints including frequent meetings, milestones, and user stories
- Expansive unit testing and GitHub automation

Notable Achievements

- First recipient of a sustainability grant from Boise State for my work on recycling 3D printer plastic, 2024
- Winner of the Idaho Health Tech Challenge, 2025
- Established memorandum of understanding between the Makerlab and a local rehab hospital, 2025
 - A result of my work creating custom equipment for individuals with physical impairments by means of 3D modeling & printing, CNC machining, vacuum forming, and consistent communication with patients.
- Member of the Student Philanthropy Board, 2025